Hierarchical Modeling

SNU Computer Science & Engineering

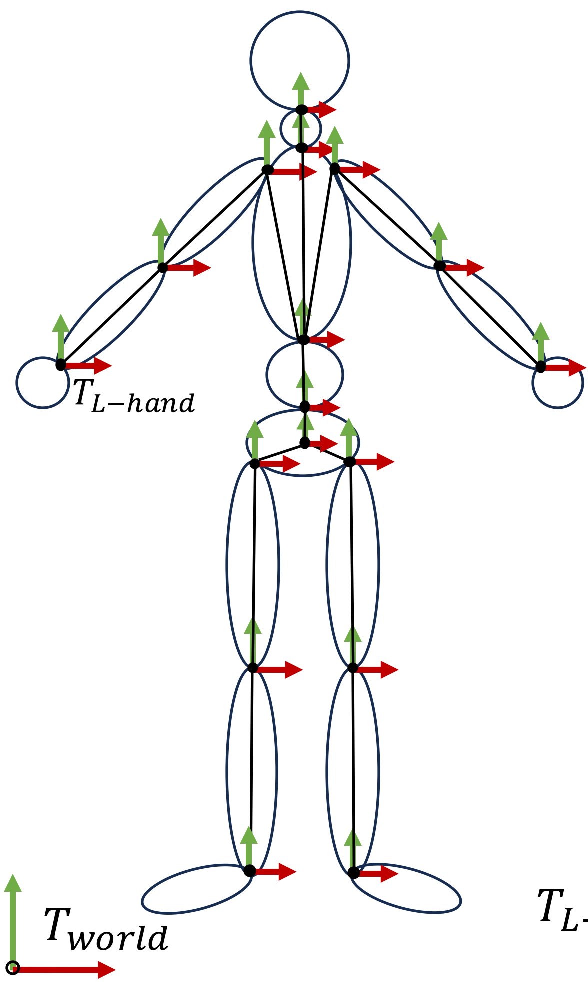
Some materials are taken from the lecture slides of MIT Computer Graphics courses taught by Justin Solomon

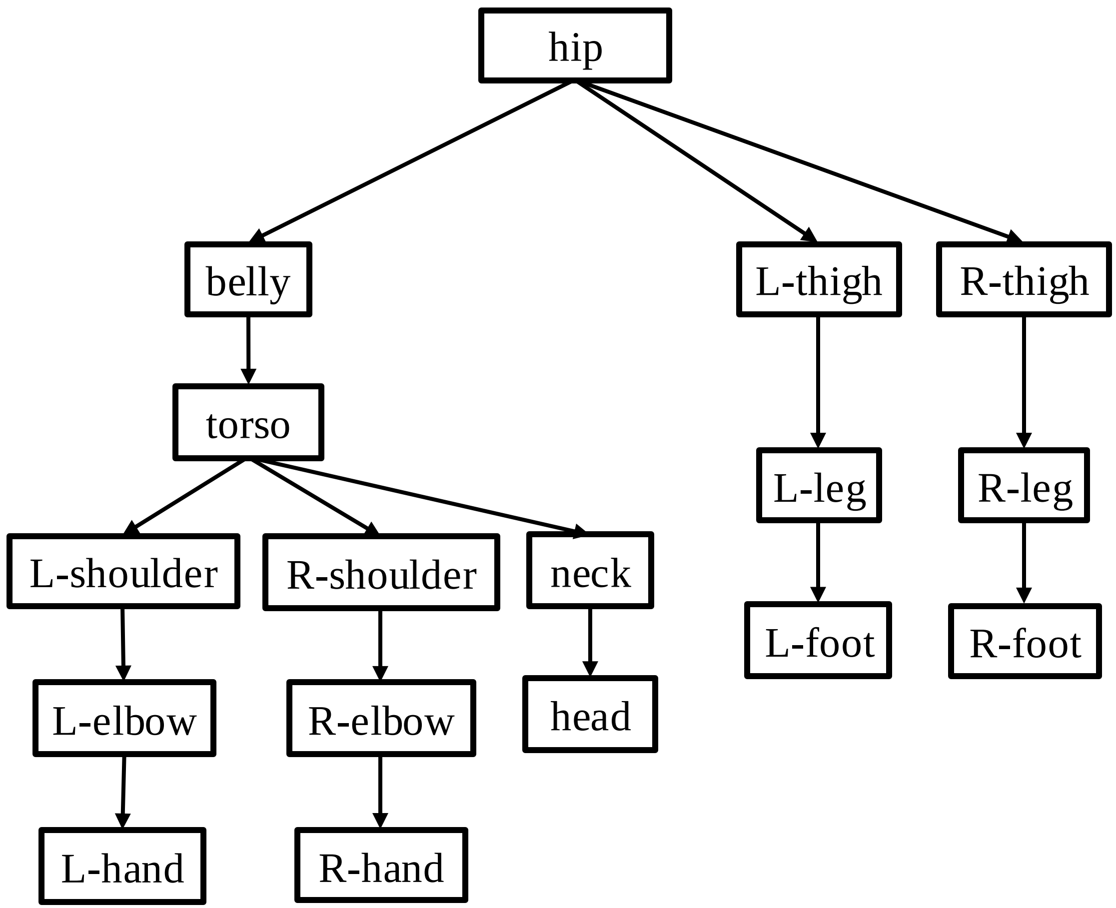
Hierarchical Modeling

- ***Complex objects*** are typically composed of ***many simple objects***. For example, a car is approximately made up of 30,000 parts and each part is even complex enough

- ***Hierarchical modeling*** creates complex objects by combining simple primitives (e.g., triangles, box, cylinders, spheres) into complex aggregates

Hierarchical Modeling: Example





$$T_{L-hand} = T_{hip} \cdot T_{belly} \cdot T_{torso} \cdot T_{L-shoulder} \cdot T_{L-elbow} \cdot T_{L-hand}$$

Static vs. Dynamic Models

- *Static* Hierarchical Model
  - Relative transformations do not change

- *Dynamic* Hierarchical Model
  - Relative transformations may change on the fly (i.e., the model is animatable)
  - The movement between connected components are often described separately by introducing ***joints***

Kinematics

- ***Kinematics*** is the study of motion of systems of bodies (i.e., hierarchical models) without considering the physical forces acting on them

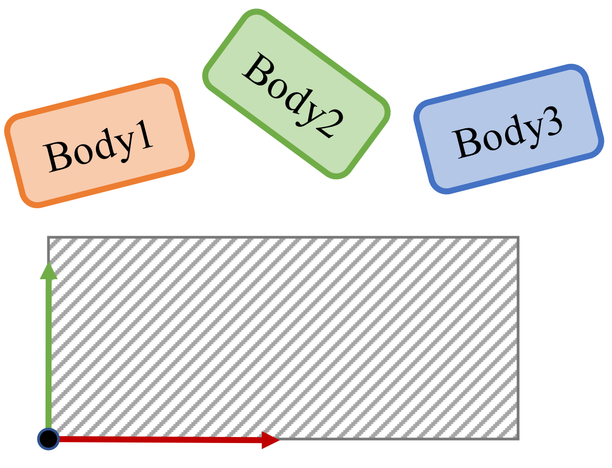
- It was developed in classical mechanics to describe motion of complex objects

- A tree structure of **joints** and **bodies** (links) is used, where the base (root) link can be chosen arbitrarily

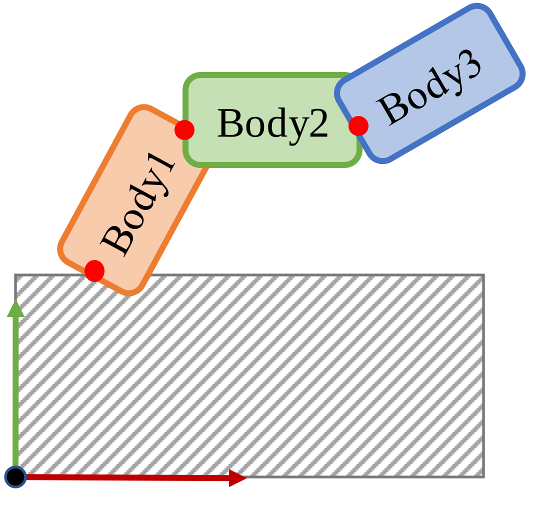
What Is a Joint?

- A joint is an area of connection where two or more bodies meet, allowing relative movement

- Mathematically, joints are constraints that reduce the ***degrees of freedom*** of a system



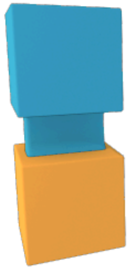
(3+3+3) DoFs



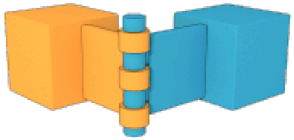
(1+1+1) DoFs

Types of Mathematical Joints

- A ***fixed (weld)*** joint does not allow any movement between connected links (0-DoF)



- A ***revolute (hinge)*** joint allows rotation about a fixed axis (1-DoF)



- A ***prismatic (sliding)*** joint allows translation along a fixed axis (1-DoF)

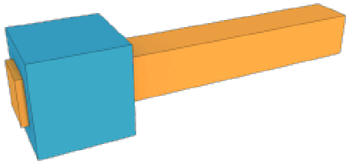
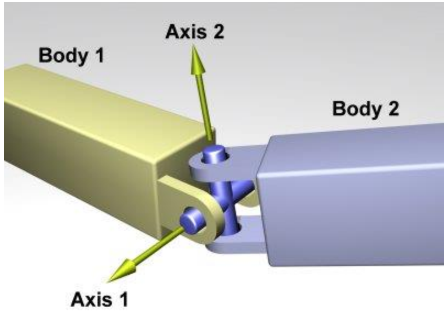


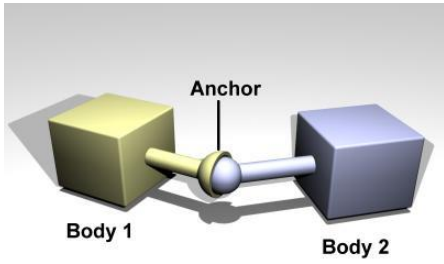
Image source: https://light-and-shadows.com/documentation/interact/

Types of Mathematical Joints

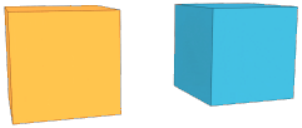
- A ***universal*** joint allows rotation around two fixed axes (2-DoFs)



- A ***ball-and-socket*** joint allows rotation around an arbitrary axis (3-DoFs)



- A ***free*** joint can rotate and translate freely in space (6-DoFs)



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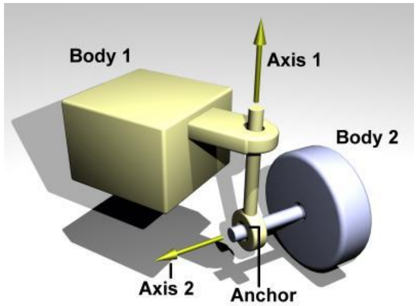
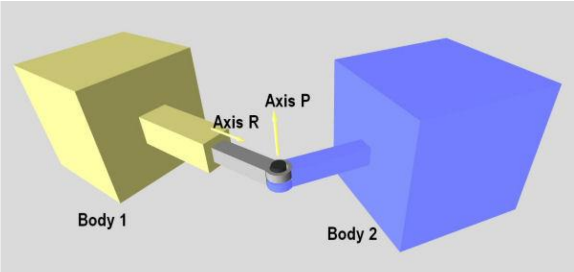
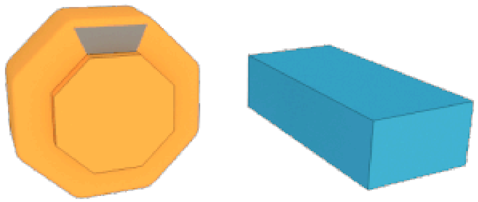
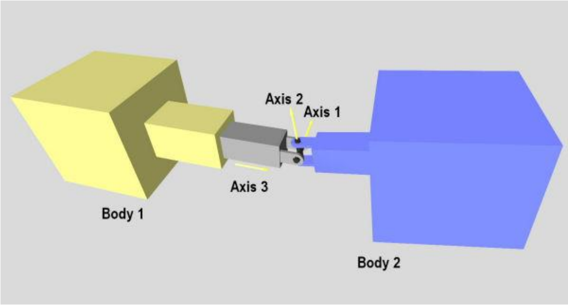
Types of Mathematical Joints

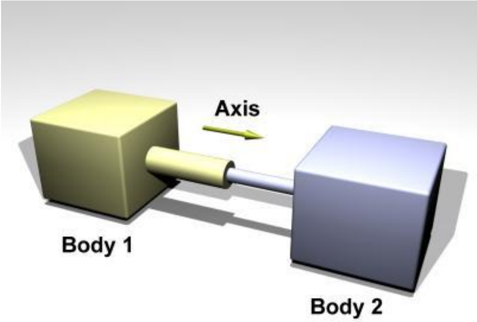
- And many more…

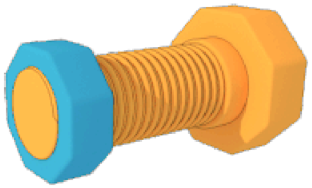












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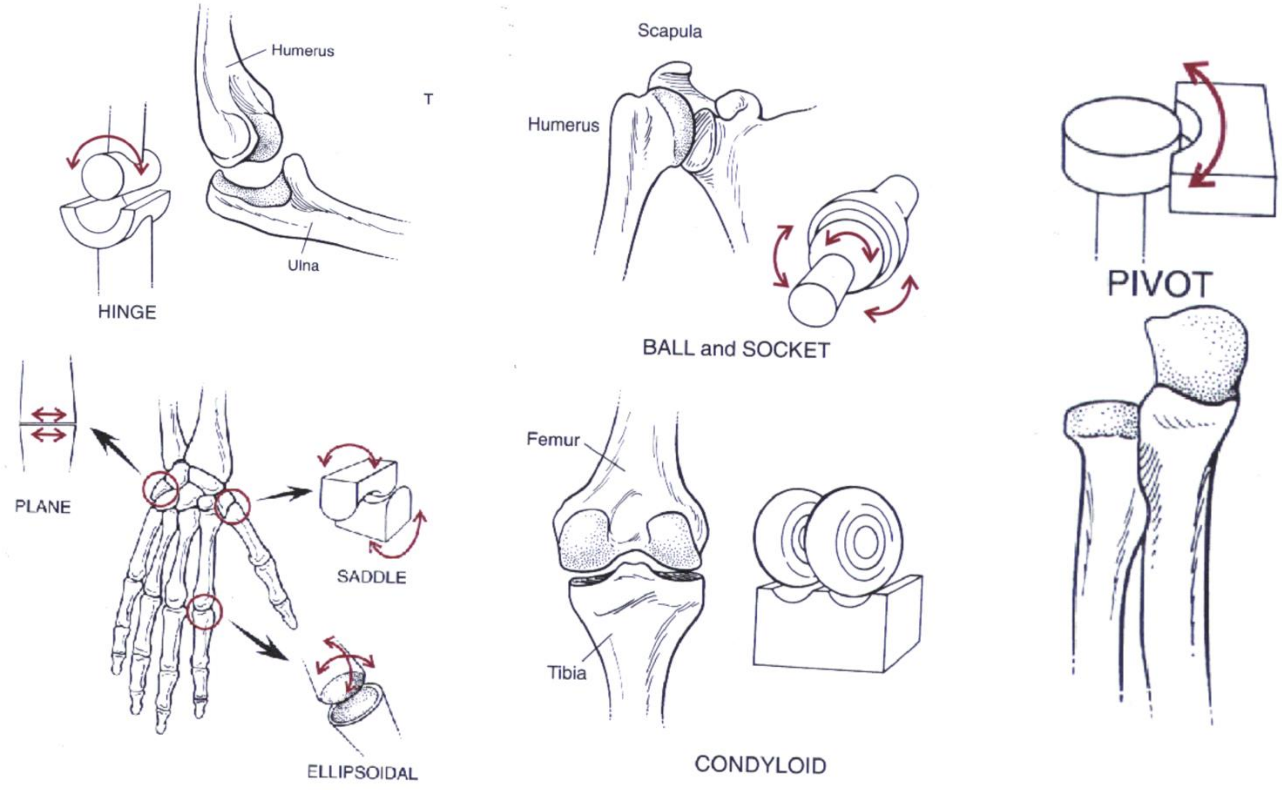
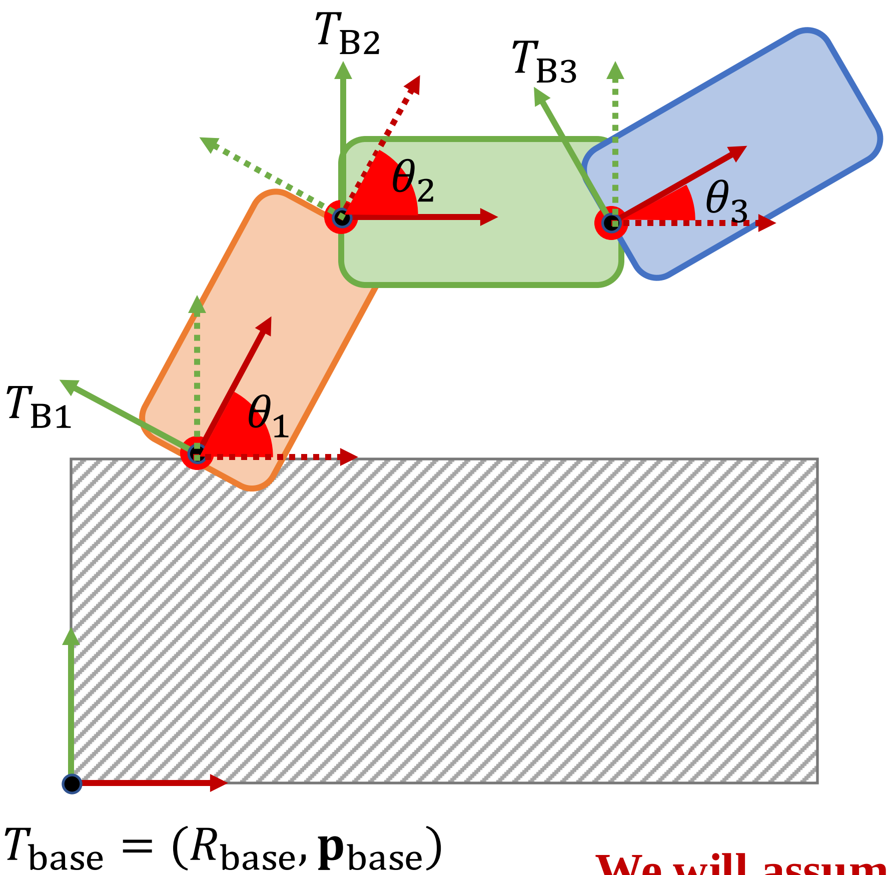
Biological Joints

- Human/Animal joints are much more complicated!



Local vs. Global Configuration



- Local (Joint) Configuration
  - Values that minimally describe the system (i.e., minimal coordinate system)
  - Joint angles $(\theta_1, \theta_2, \theta_3)$
- Global Configuration
  - Values that maximally describe the system (i.e., maximal coordinate system)
  - Rotations and translations (T_{B1}, T_{B2}, T_{B3}) of each body w.r.t. T_{base}

We will assume rigid transformation for simplicity

Forward and Inverse Kinematics

- ***Forward Kinematics*** (FK)
  - The process to compute global configuration given joint (local) configuration

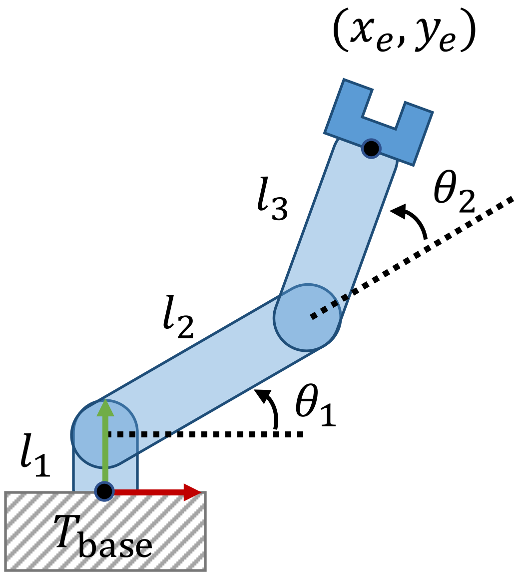
$$\underbrace{(R_i, \mathbf{p}_i)}_{\text{Unknown}} = FK_i(\underbrace{\theta_1, \theta_2, \dots, \theta_N}_{\text{Known}})$$

- ***Inverse Kinematics*** (IK)
  - The process to compute joint configuration given (often partial) global configuration

$$(\underbrace{\theta_1, \theta_2, \dots, \theta_N}_{\text{Unknown}}) = FK_i^{-1}(\underbrace{R_i, \mathbf{p}_{\boldsymbol{i}}}_{\text{known}})$$

Forward Kinematics: A Simple Example

- A simple robot arm in 2-dimensional space
  - 3 bodies, 2 revolute (hinge) joints
  - Compute the position of the end-effector (x_e, y_e) given joint angles (θ_1, θ_2) w.r.t. T_{base}



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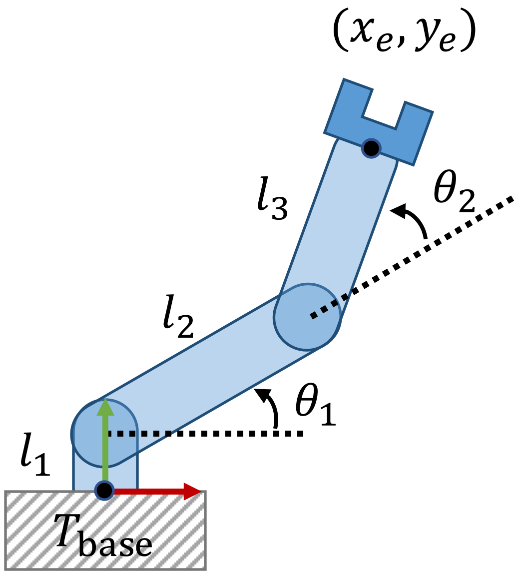
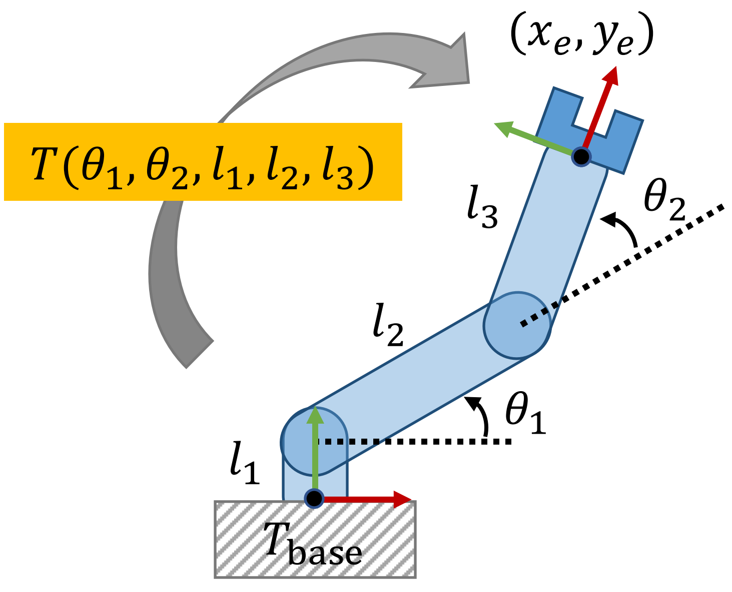


$$x_e = l_2 cos\theta_1 + l_3 \cos(\theta_1 + \theta_2)$$

$$y_e = l_1 + l_2 sin\theta_1 + l_3 \sin(\theta_1 + \theta_2)$$

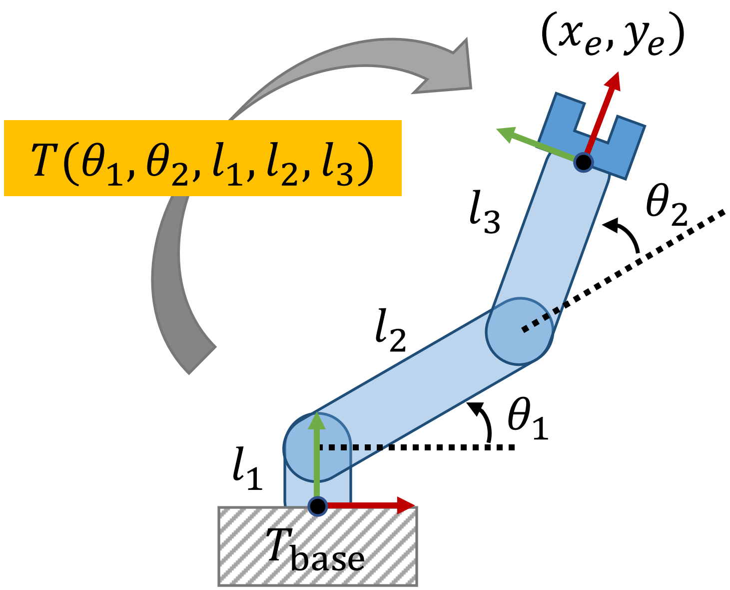
Forward Kinematics: A Simple Example

- Forward kinematics map as coordinate transformation
  - The body local coordinate system of the end-effector was initially coincided with the reference coordinate system T_{base}
  - Forward kinematics map transforms the position and orientation of the end-effector according to joint angles and link lengths



$$\begin{bmatrix} x_e \\ y_e \\ 1 \end{bmatrix} = \begin{bmatrix} T(\theta_1, \theta_2, l_1, l_2, l_3) \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

A Chain of Transformations

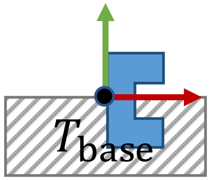


$$\begin{bmatrix} x_e \\ y_e \\ 1 \end{bmatrix} = \begin{bmatrix} & & \\ & T(\theta_1, \theta_2, l_1, l_2, l_3) & \\ & & \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$T = \mathrm{T_y}(l_1) R(\theta_1) \mathrm{T_x}(l_2) R(\theta_2) \mathrm{T_x}(l_3)$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & l_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} cos\theta_1 & -sin\theta_1 & 0 \\ sin\theta_1 & cos\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & l_2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} cos\theta_2 & -sin\theta_2 & 0 \\ sin\theta_2 & cos\theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & l_3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Interpretation (Left-to-Right)



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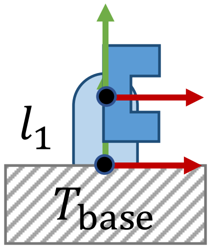

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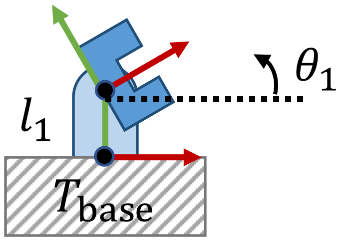

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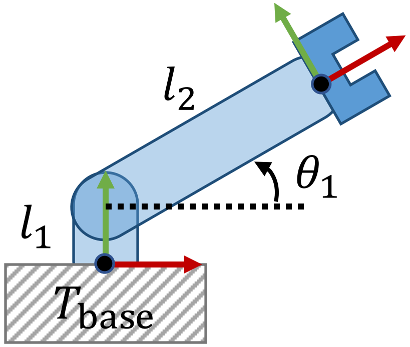

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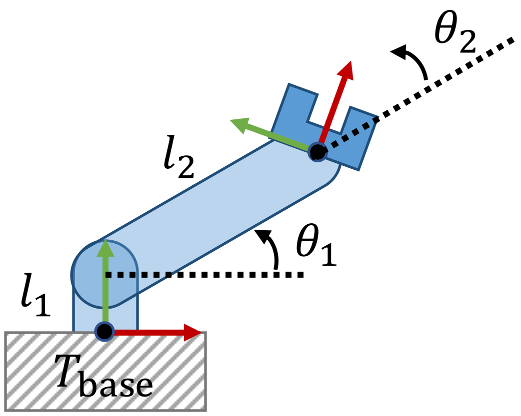

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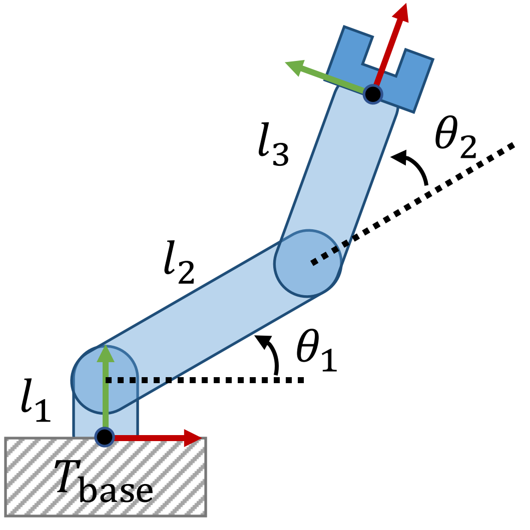

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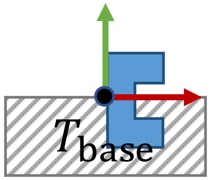
Interpretation (Right-to-Left)



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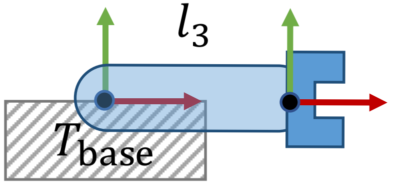

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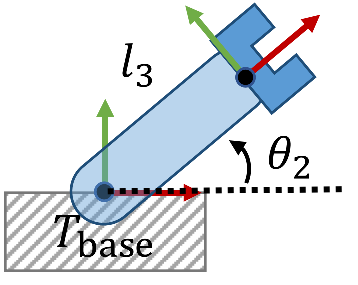

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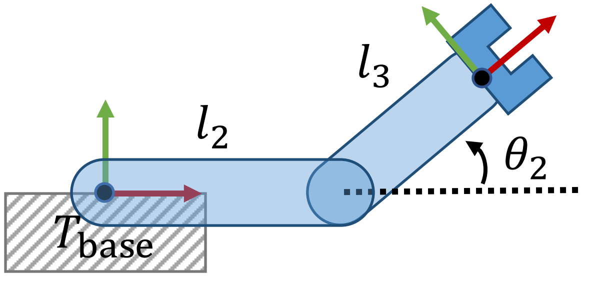

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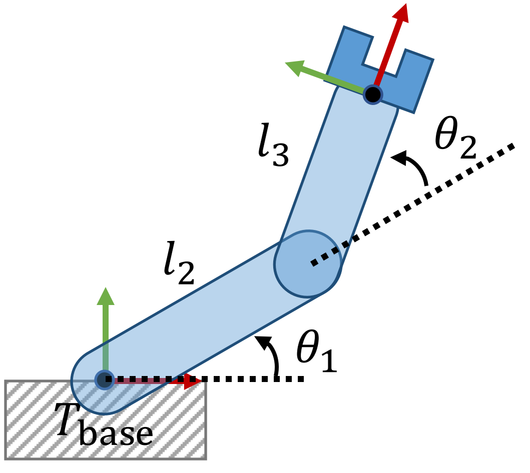

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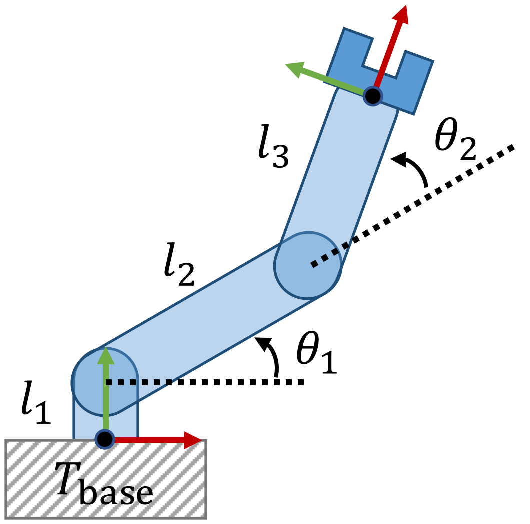

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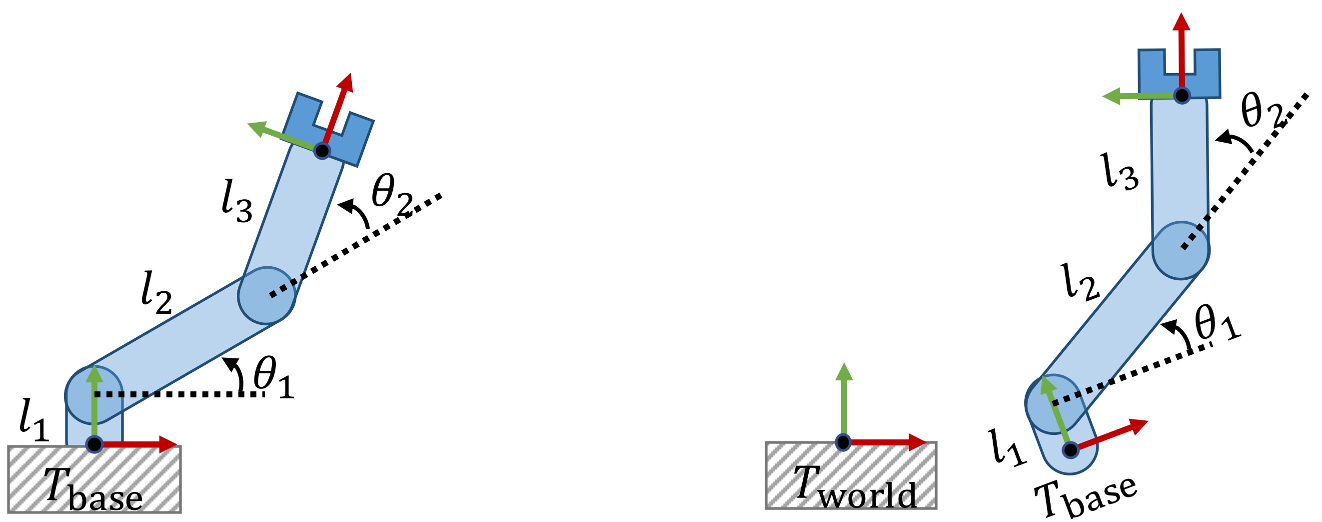
Joint and Link Transformations

- In general, a FK map can be represented as follows

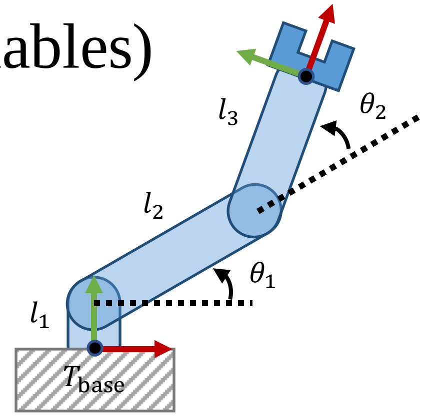
$$T = T_{\text{base}} T_{L1} T_{J1} T_{L2} T_{J2} \cdots$$

- In the previous example, T_{base} was ignored because it coincides with the world frame. If the base can move freely (e.g., the right figure), it should also be considered



Joint and Link Transformations

- ***Link*** transformations
  - Defines a frame relative to its parent (often fixed)
- ***Joint*** transformations
  - Represent joint movement (often set by variables)



$$T = \mathrm{T_y}(l_1)R(\theta_1)\mathrm{T_x}(l_2)R(\theta_2)\mathrm{T_x}(l_3)$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & l_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} cos\theta_1 & -sin\theta_1 & 0 \\ sin\theta_1 & cos\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & l_2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} cos\theta_2 & -sin\theta_2 & 0 \\ sin\theta_2 & cos\theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & l_3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

How to Represent a Floating Base?

- A ***free*** joint (rotation + translation) is used

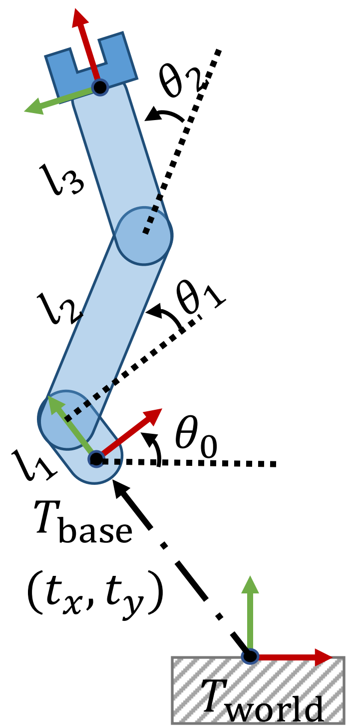
$$T = T_x(t_x)T_y(t_t)R(\theta)$$ (in 2D space)

$$= \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T = T_x(t_x)T_y\left(t_y\right)T_z(t_z)R_x(\theta)R_y(\psi)R_z(\phi)$$ (in 3D space)

$$= \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\psi & 0 & \sin\psi & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\psi & 0 & \cos\psi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\phi & -\sin\phi & 0 & 0 \\ \sin\phi & \cos\phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

How to Represent a Floating Base?



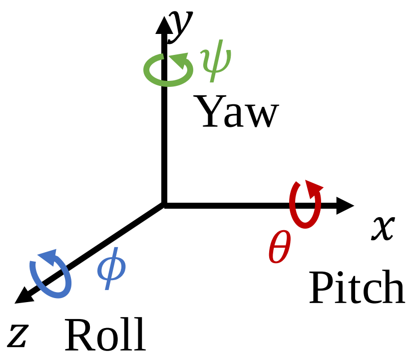
$$T = \underbrace{T_x(t_x)T_y(t_y)R(\theta_0)}_{T_{\text{base}}}\underbrace{\mathrm{T_y}(l_1)}_{T_{L1}}\underbrace{R(\theta_1)}_{T_{J1}}\underbrace{\mathrm{T_x}(l_2)}_{T_{L2}}\underbrace{R(\theta_2)}_{T_{J2}}\underbrace{\mathrm{T_x}(l_3)}_{T_{L3}}$$

How to Handle Ball-and-Socket Joints?

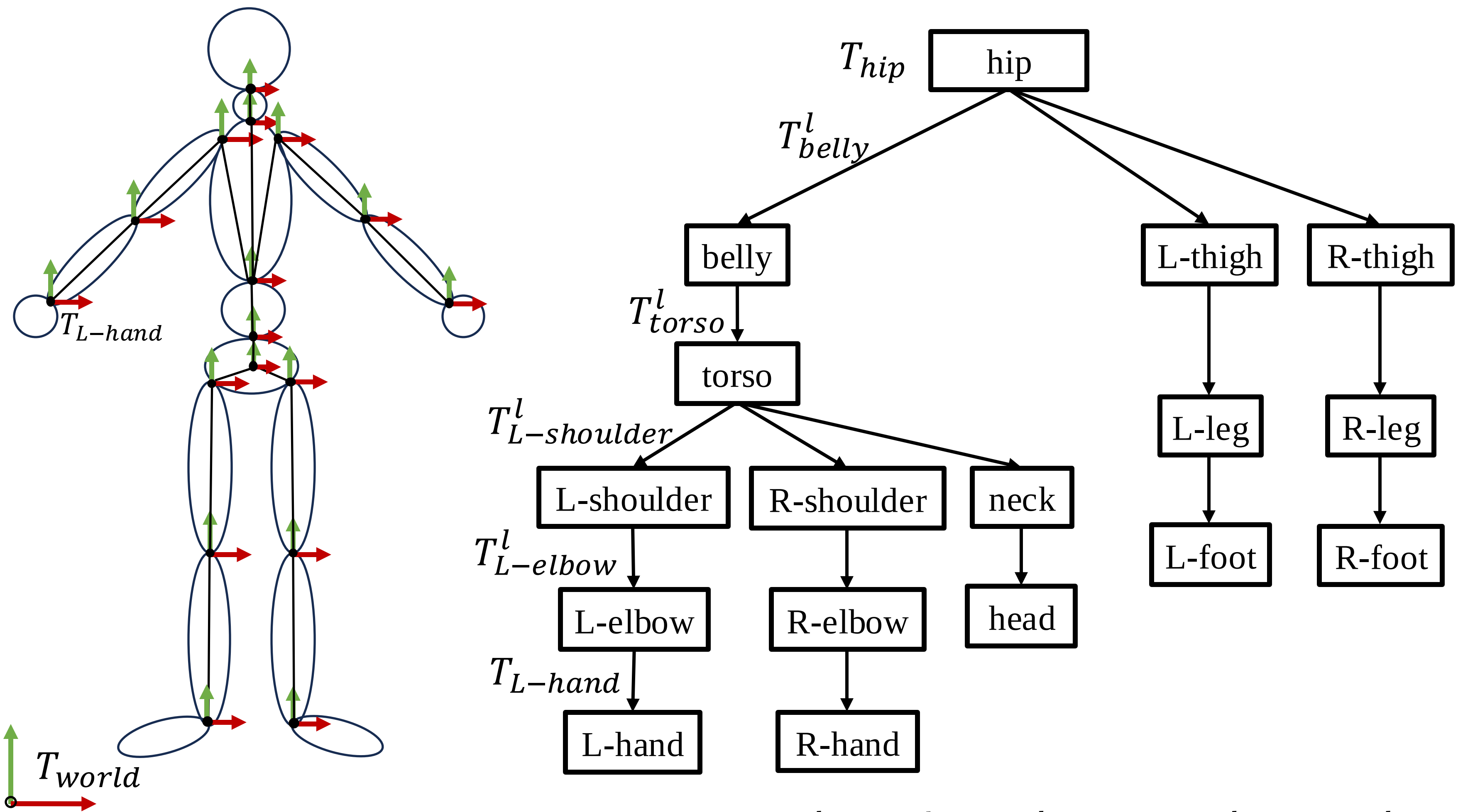
- Three revolute joints whose axes intersect at a point (equivalent to Euler angles)

$$T = R_x(\theta)R_y(\psi)R_z(\phi)$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\psi & 0 & \sin\psi & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\psi & 0 & \cos\psi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\phi & -\sin\phi & 0 & 0 \\ \sin\phi & \cos\phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

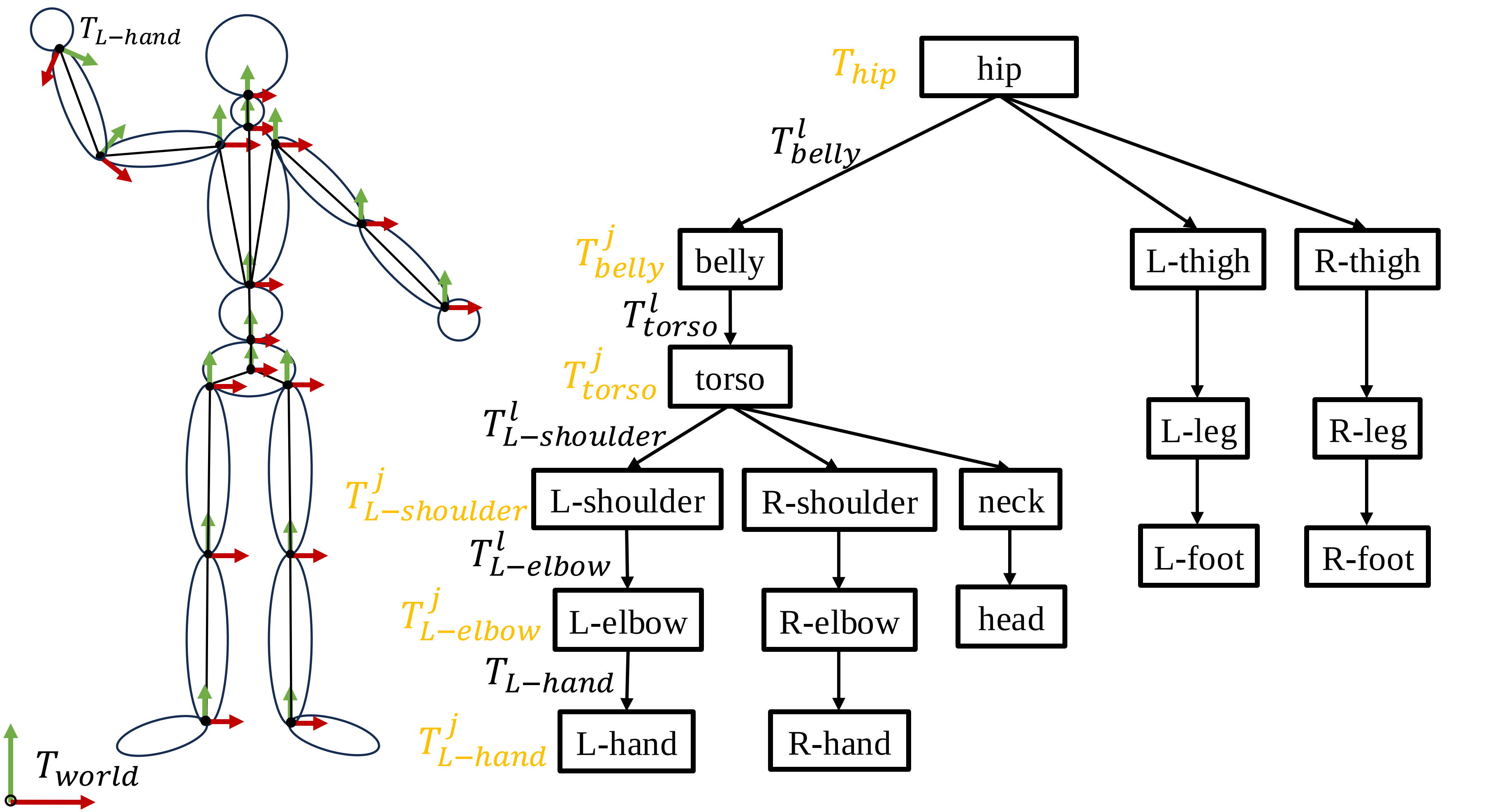


Hierarchical Modeling: An Example



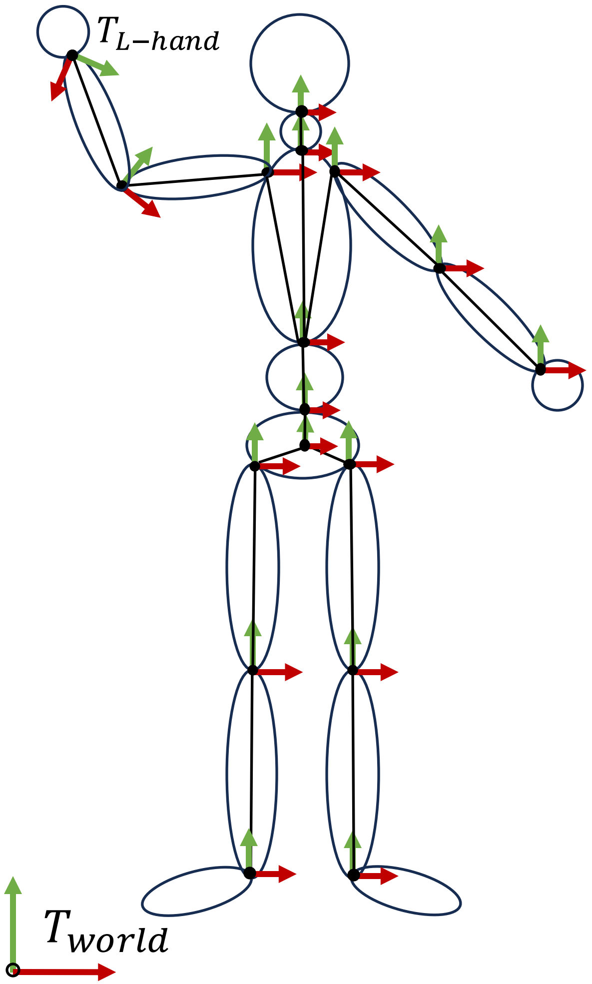
$$T_{L-hand} = T_{hip} \cdot T^l_{belly} \cdot T^l_{torso} \cdot T^l_{L-shoulder} \cdot T^l_{L-elbow} \cdot T^l_{L-hand}$$

Hierarchical Modeling: An Example



$$T_{L-hand} = T_{hip} \cdot T^l_{belly} \cdot T^j_{belly} \cdot T^l_{torso} \cdot T^j_{torso} \cdot T^l_{L-shoulder} \cdot T^j_{L-shoulder} \cdot T^l_{L-elbow} \cdot T^j_{L-elbow} \cdot T^l_{L-hand} \cdot T^j_{L-hand}$$

Hierarchical Modeling: An Example



- Each node may include:
  - Link transf. (w.r.t. its parent)
  - Joint transf. (w.r.t. its link transf.)
  - A list of child joints
  - A list of visual node
    - Geometry
    - Geometry transf. w.r.t. joint transf.
    - Color
    - Texture

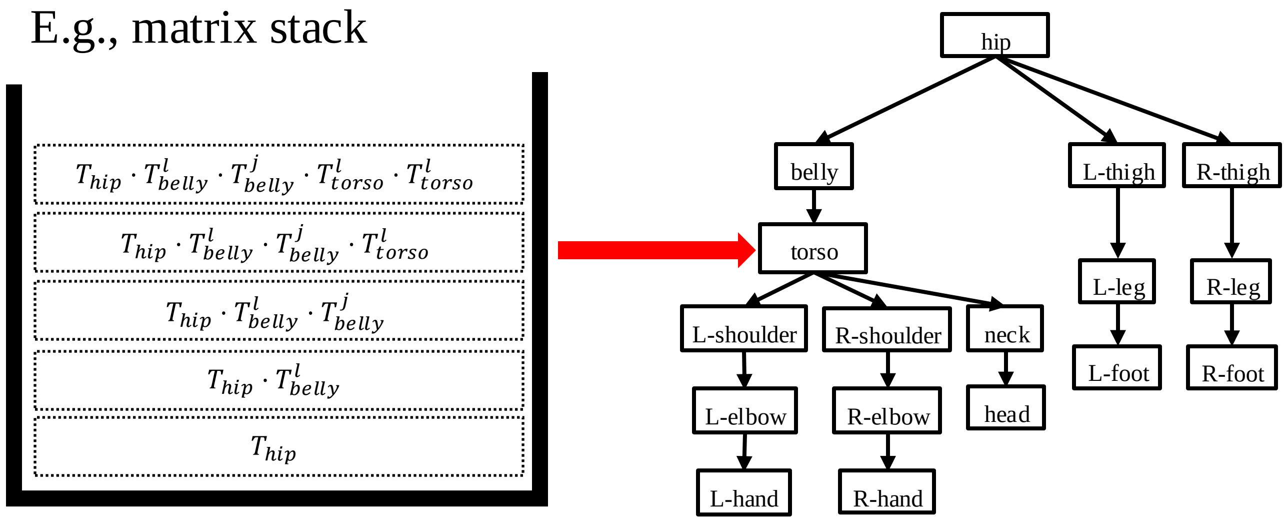
Processing Kinematic Trees

- To process a specific node, all the transformations along the path from the base (root) node are required
- ***Depth-first*** traversal
  - Visit a node, then visit subtrees
  - When visiting each node, apply a new state to the previous state
  - When returning back to its parent, restore its state to the previous state
  - **How can we implement this?**
    - It would be a bad idea to use the inverse matrix when returning back, why?

Processing Kinematic Trees

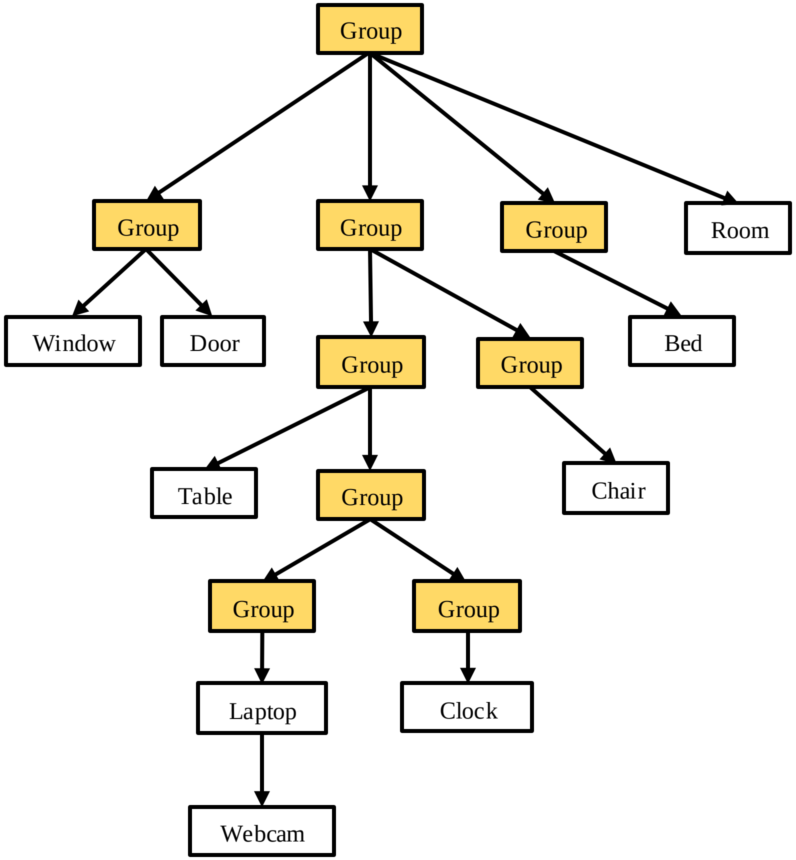
- A ***state stack*** can be useful during traversal, which is also implemented in OpenGL (deprecated in modern OpenGL)
  - ***Push*** the current state when visiting a new node
  - ***Pop*** the current state when returning back

E.g., matrix stack



Scene Graphs

- The same idea can also apply to the entire scene



Summary

- Kinematics is the study of motion of complex objects with articulation
- Kinematics does not consider physics (forces, mass)
- Forward kinematics is straightforward
  - Forward kinematics map is just a series of coordinate transformations (a series of matrix multiplication)
- Inverse kinematics (not covered in this lecture) may not be trivial
  - There exist many or no solution in many cases
- A scene graph is a generalization of kinematics while considering properties other than motion